

DAY 2

Array

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★ Insert / delete in middle needs Shifting

• Initial :

$a = [10, 20, 30, 40, 0, 0, 0, 0]$, Size = 4

① - Insert 15 at Index 1

Before :-

$[10, 20, 30, 40, 0, 0, 0, 0]$
0 1 2 3 4 5 6 7

- Shift right

40 → index 4

30 → index 3

20 → index 2

- After $a[1] = 15$ put at Index array will be:

$[10, 15, 20, 30, 40, 0, 0, 0]$, Shift = 3, Size = 5
0 1 2 3 4 5 6 7

② - Insert 50 at End

Before :- $[10, 15, 20, 30, 40, 0, 0, 0]$
0 1 2 3 4 5 6 7

then after put 50 at 5 index array will be:

0 1 2 3 4 5 6 7
 $[10, 15, 20, 30, 40, 50, 0, 0]$, Shift = 0, Size = 6

③ Delete index 0

Delete 10

- Shift left :

15 → 0 index

20 → 1 index

30 → 2 index

40 → 3 index

50 → 4 index

Set last position to 0 :

Shifts = 5, size = 8

[15, 20, 80, 40, 50, 0, 0, 0]
 0 1 2 3 4 5 6 7

$A = \text{arr}$, $[0, 0, 0, 0, 0, 0, 0, 0] = 0$

1 2 3 4 5 6 7
0 0 0 0 0 0 0

Shift right
 0 shifts
 1 shift
 2 shifts
 3 shifts

After $A[7] = 15$ put at index array

Shift = 3, size = 8
 [10, 12, 20, 30, 40, 0, 0, 0]
 0 1 2 3 4 5 6 7

Before : [10, 12, 20, 30, 40, 0, 0, 0]
 0 1 2 3 4 5 6 7

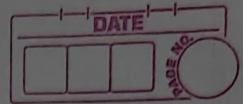
then after put 15 at index array

[10, 12, 20, 30, 40, 20, 0, 0]
 0 1 2 3 4 5 6 7

Shift left :
 0 shifts
 1 shift

Shift right :
 0 shifts
 1 shift
 2 shifts

Stack Using array



Suppose inputs are :

Capacity 3

push(10), push(20), pop(), peek(), display()

- Initially :-

Stack = [_, _, _], top = -1

		2
		1
		0
top →		-1

1. push(10)

Stack[top] = 10 So top is now 0

Stack = [10, _, _], top = 0

		2
		1
top →	10	0
		-1

2. push(20)

top = 1 and at index 1, 20 is added
So,

Stack = [10, 20, _], top = 1

		2
top →	20	1
	10	0
		-1

3. push(30)

top = 2, Stack[2] = 30

So,

Stack = [10, 20, 30], top = 2

top →	30	2
	20	1
	10	0
		-1

4. pop()

return Stack[top--]

Return 30, then top becomes 1.

Stack = [10, 20, 30], top = 1

		2
→	20	1
	10	0
		-1

peek()

return stack[top]

So

stack[1] = 20

Now top element is 20 with top at 1

top →	20	2
	10	1
		0

Display()

loop runs :-

i = 1

stack[i] ⇒ 20, i--

i = 0

stack[0] ⇒ 10, i--

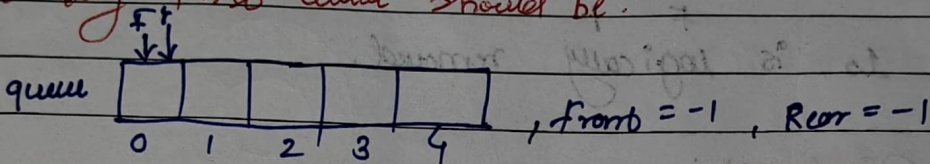
Stack elements (top to bottom): 20 10

Queue Using Array

Suppose inputs are:-

- Enqueue(10)
- Enqueue(20)
- Enqueue(30)
- Dequeue()
- Display()

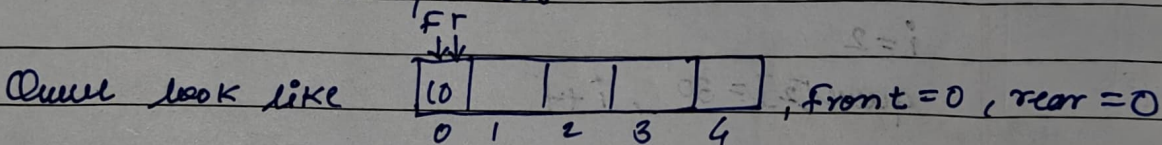
At Very first Queue should be:-



1. Enqueue 10

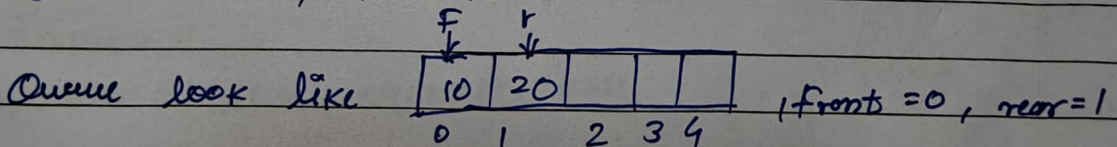
So, front = -1 :

front = 0, rear = 0, queue[0] = 10



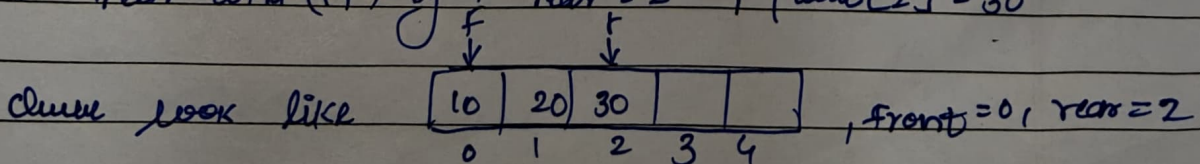
2. Enqueue 20

So, rear = 1, queue[1] = 20



3. Enqueue 30

So rear will ++ by 1 rear = 2, queue[2] = 30



4. Dequeue :-
 $queue[front++]$ which will be stored in removed

So,
 $removed = queue[0] = 10$, $front = 1$

Queue will look like

0	1	2	3	4
10	20	30	.	.

10 is logically removed.

5. Display :-

$i = 1$

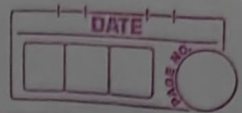
$queue[i] = 20$, $i++$

$i = 2$

$queue[i] = 30$, $i++$

Queue elements : 20 30

Circular Queue



Suppose operations:

Enqueue(10)

Enqueue(20)

Enqueue(30)

Dequeue

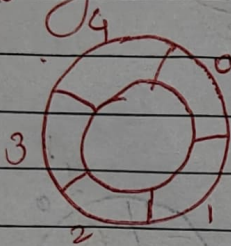
Enqueue(40)

Enqueue(50)

Enqueue(60)

display()

Initially the queue:-



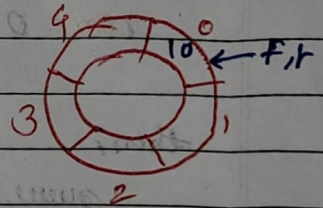
front = -1

rear = -1

1. Enqueue 10

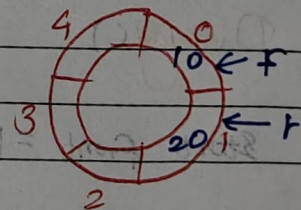
queue is empty:

front = rear = 0, queue[0] = 10



2. Enqueue 20

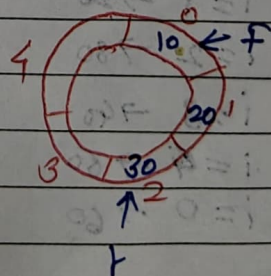
rear++, r = 1, queue[1] = 20



3. Enqueue 30

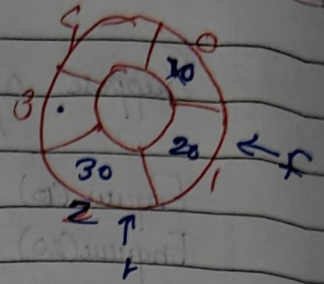
rear = 2

queue[2] = 30

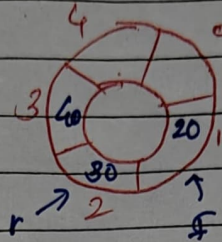


4. Dequeue ¹⁰
removed = queue[0]

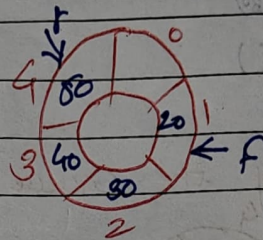
So,
removed = 10, front++, front = 1



5. Enqueue (40)
rear = 3, queue[3] = 40



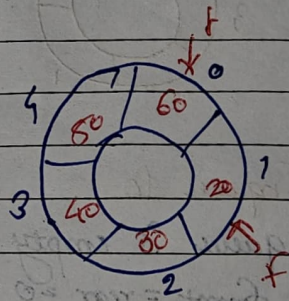
6. Enqueue (50)
rear = 4, queue[4] = 50



7. Enqueue (60)
rear = 2 max - 1, So circularly
rear = 0,

then;

queue[0] = 60



8. Display()

Start front = 1

i = 1 → 20

i = 2 → 30

i = 3 → 40

i = 4 → 50

i = 0 → 60

Circular queue elements:
20 30 40 50 60

Balanced Brackets

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Suppose String = "{[()]}"

Character	Operation	Stack
{	push {	[{]
[	push [	[{, [
(	push (	[{, [, (
)	pop (, match	[{, [
]	pop [, match	[{]
}	pop {, match	[]

Finally: `stack.isEmpty() = true`

O/P: "{[()]} → Balanced